

COMPUTER SCIENCE TRIPOS Part IB – 2024 – Paper 7

10 Further Human–Computer Interaction (afb21)

This question relates to the design of an online tool for use by a local radio station, that can be used to more efficiently generate scripts for their bulletins. The station manager has asked you to investigate whether presenters could ask ChatGPT to automatically generate the evening weather report, which typically includes a summary of today’s weather, and forecasts for tomorrow and the rest of the week.

Designing smart systems

- (a) Sketch a user interface that might be used as the front end to a generative AI large language model, as an alternative to the standard conversational style of interaction with ChatGPT. [3 marks]

Answer: The sketch should minimally present weather data, some consideration of customisation / variation, and controls to generate, preview and adjust.

Design of visual displays

- (b) Add labels to refer to two specific parts of your sketch, and explain how the graphical properties of those elements correspond to their function within your design. [4 marks]

Answer: Any reference to the chosen graphical properties is acceptable, if there is a description of mapping, depictive symbols, topological relations etc.

Designing efficient systems

- (c) What analytic method could be used to predict whether your alternative design would be more time-efficient than the standard style of interaction with ChatGPT? [1 mark]

Answer: The Keystroke Level Model can be used to compare time taken to generate text when using each interface.

Evaluating interactive system designs

- (d) Based on the analytic method that you suggest, provide calculations for an estimated (within an order of magnitude) comparison of (i) the station manager’s suggestion to use ChatGPT, and (ii) the use of the user interface you have sketched, in a typical usage. [4 marks]

Answer: In the station manager’s approach, it will be necessary to type instructions into the ChatGPT prompt. The preface might be “Generate a weather report for a radio station bulletin”, followed by individual sections, such as “the weather today was sunny, and the temperature high was”, after which it will be necessary for the presenter to look up the temperature and either transcribe or copy and paste into the prompt window. This will need to be repeated for each of the three elements of the weather report. In addition to temperature, there might be information on wind and precipitation. I estimate that each of these elements would involve about 50 keystrokes, and maybe 10 mouse navigation and scrolling actions. Assuming 0.5s per keystroke, and 1s per mouse action, my rough estimate for use of ChatGPT is $((50 \times 0.5) + 10) \times 10$, or 350 seconds - between 5 and 6 minutes.

In my alternative approach, the weather data has been preloaded from an online data service, and the presenter simply has to select the options for customisation prompts that will add some entertainment and variability to the weather report. I estimate that the audience will

not care if they hear an identical report after 25 days, so the presenter should have at least 5 options for each of 2 customisations, to be combined in any order. The time required to select one of five customisations from a drop-down menu will be about 1 second, followed by clicking the generate button. I estimate that the text will be generated within about 3-5 seconds, which would be two orders of magnitude faster than standard interaction with ChatGPT.

Evaluating
interactive system
designs

- (e) Describe an empirical approach that could be used to evaluate whether the estimates that you have calculated are justified. [4 marks]

Answer: These estimates could be tested using a controlled experiment, in which conversational interaction with ChatGPT was compared to a simple web interface providing the planned controls. The experiment could be carried out more cheaply by using a prototype that displays predefined data rather than implementing a fully functional back-end. A sample would be recruited of station employees representative of the presenters who might use the system in future. Each person would be asked to generate a weather report using first one method, then the other, with half using ChatGPT first. The times taken are recorded with a stopwatch. The sample mean times using each method should then be compared for statistical significance. A paired-samples t-test would be appropriate. It seems likely that a two order of magnitude difference will be found significant at the 5 percent level, even with a small sample.

Designing
meaningful /
complex systems

- (f) Describe up to four further considerations, beyond time-efficiency, that might be relevant to the station manager's decision to deploy such a tool. Make specific reference to topics from the HCI literature, e.g. from the lectures, notes or readings, but not media reports of AI. [4 marks]

Answer: Firstly, it is possible that measurement of the intervention will be subject to the Hawthorne effect, such that any change appears potentially profitable.

Second, we might consider the cognitive dimension of viscosity - if it was necessary to make changes to the resulting script, how would this be achieved?

Third, are there any problems of bias resulting from the interaction design, such as the availability heuristic meaning that presenters no longer consider reporting on any option that is not included in the interface?

Fourth, could systems like this make a positive contribution to the wicked problem of climate change, by ensuring that local weather reports make connections to local mitigations and adaptations that would be beneficial on a global scale?
